

DOLITRAIL · REQUIREMENTS

DoliTrail requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/dolitrail		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-03	Amish Chadha	TRL 3 status from DLT-CAL-001 on the constructable design; R10 and R11 added (DLT-DDR-001); targets R1 to R9 unchanged
0.1	2026-09-30	Amish Chadha	Initial scaffold

DoliTrail requirements

Requirements for the kit fitted to the reference doli (two 60 mm bamboo poles, 2.55 m long, at 550 mm centres) carrying the rated patient. R1 to R9 keep the targets set at TRL 1; R10 and R11 were added at TRL 2 (DLT-DDR-001). Status is the paper result at TRL 3 from DLT-CAL-001; every requirement is verified by test at TRL 4. Requirements not met or at risk are posed to Amish as decisions in docs/REVIEW.md and docs/06-design-decisions.md .

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 3 or later)	TRL 3 status
R1	Fits real dolis	Clamps grip poles of 40 to 80 mm diameter and hold without slipping under a 1.5 kN load (target)	Fit trials on 10 field-made dolis and CalRig pull tests	Met on paper: the clamps close on 40 and 80 mm poles in the model; two clamps on one pole resist 3.96 kN wet (factor 2.6); bamboo crushing factor 2.0 to confirm on real poles
R2	Quick to fit	Two people fit the kit to a doli in under 5 minutes	Timed trials using only the fitting card	Met on paper (estimate): about 4.7 min, with the wheel carried fitted in the fork unit
R3	Carry load	Rated for a 120 kg (265 lb) patient plus doli, with a 2 to 1 safety margin on the wheel, fork and clamps	CalRig proof-load test	At risk: fork, arms and clamps met (lowest factor 2.02 on yield at twice the rated load); the wheel and tyre are not catalogue-rated for twice the wheel load. Decision for Amish
R4	Reduce bearer load	Average vertical load per bearer reduced by at least 60 per cent on level trail compared with carrying	Load cells in the harness cups during trail trials	Met on paper (estimate): 79 per cent (about 14 kg per bearer against 65 kg)
R5	Brake holds on descents	Holds the rated load stationary on a 30 per cent grade, wet	Wet ramp test	At risk: brake factor 1.23 and tyre grip factor 1.44 on a wet test ramp, but 0.84 on wet clay or mud. Decision for Amish
R6	Narrow trails	Overall width with wheel at or below 700 mm (28 in)	Measure; walk a marked 0.8 m trail course	Met on the reference doli: 680 mm; any doli with poles at more than 570 mm centres is wider than 700 mm by itself
R7	Portable	Kit mass at or below 10 kg (22 lb) and carried by one person	Weigh; timed carry of the bagged kit	Not met: 14.8 kg in the bag. Decision for Amish
R8	Lift over obstacles	Two bearers lift the fitted doli over a 400 mm step without removing the wheel	Obstacle course trial	Not met with two bearers: the poles must rise 450 mm to clear the step, about 71 kg each to chest height. Decision for Amish
R9	Low cost	Parts cost at or below \$1,500 USD for the kit	Costed bill of materials	USD 455: under the value-engineering target of USD 1,500 by USD 1,045
R10	Patient release	Each patient strap frees with one pull of one hand in 3 s or less	Timed trial, gloved hand	Met by design: cam buckles with a pull tab
R11	Brake within reach	The front bearer works the brake and its parking lock without letting go of the poles	Trial with bearers of 1.5 to 1.8 m	Met on paper: lever on the front right pole about 100 to 150 mm behind the pole loop

■ Assumptions

- Rated load: a 120 kg patient on a 10 kg doli; with the 12.3 kg fitted kit the rolling weight is 142 kg (DLT-CAL-001, A1).
- The wheel is placed under the patient's hips to within 100 mm; bearers steady the doli with about 5 per cent of the rolling weight each.

- Typical routes roll on level ground and moderate grades; bearers lift and carry at steps, rocks and water.
- Bamboo poles carry the patient through four clamps at about 600 mm spacing near mid-span.
- Ambulance services and health committees would keep and maintain kits.

Safety: These requirements describe equipment that carries a person over dangerous ground. A paper result is not a test. No patient is carried until the TRL 4 proof loads and first checks in DLT-BLD-001 pass.